



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2012/2013

FIRST YEAR FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE (GEOGRAPHY & NATURAL RESOURCES MANAGEMENT) AND BACHELOR OF EDUCATION WITH INFORMATION TECHNOLOGY (CITY CAMPUS - SB)

NGE 101: INTRODUCTION TO GEOGRAPHY

Date: 28th August, 2013

Time: 2.00 – 4.00 p.m.

INSTRUCTIONS:

- ◆ Answer Question ONE and any other TWO (2) questions.
- ◆ Sketch maps and diagrams should be used whenever appropriate.



NGE 101 INTRODUCTION TO GEOGRAPHY

1. Examine the evolution of the discipline geography and schematically outline its subfields and interrelated disciplines and their relevance to socio-economic development. (30 marks)
2. Explain how the four (4) traditions of geography represent unifying or dividing approaches to geographic understanding. (20 marks)
3. a) Describe the four (4) geographic concepts. (10 marks)
b) Explain the importance of studying geography in today's world. (10 marks)
4. Examine factors responsible for the global climate change. (20 marks)
5. Examine the place of maps in the geographic traditions. (20 marks)
6. Discuss the relevance of geography in understanding regional integration in Kenya. (20 marks)



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(CITY CAMPUS - SB)

NGE 102: MAP INTERPRETATION

Date: 30th August, 2013

Time: 2.00 – 4.00 p.m.

INSTRUCTIONS:

- ◆ Answer Question ONE and any other TWO (2) questions.
- ◆ Sketch maps and diagrams should be used whenever appropriate.



NGE 102 MAP INTERPRETATION

1. Use the map extract Longonot Sheet 133/4 Scale 1: 50, 000 to answer the following questions.
 - a. Comment on the relief of the area covered by the map. (3 marks)
 - b. Identify two (2) natural resources and explain their importance to the economic development of the country. (6 marks)
 - c. Identify the predominant drainage pattern in the area covered and describe its associated features. (3 marks)
 - d. (i) Calculate the approximate area of Lake Naivasha in hectares. (5 marks)
(ii) Calculate the distance in kilometres of the feature shown between grid references 156170 and 217020. (3 marks)
 - e. Identify three (3) features within the border of this map. (3 marks)
 - f. Explain the factors that threaten the existence of Lake Naivasha. (6 marks)
2. a. Distinguish between Topographical and Thematic Maps. (4 marks)
b. Discuss the relevance of topographical maps to geographic studies. (15 marks)
3. a. Explain the rationale of reduction and enlargement in mapwork. (10 marks)
b. Discuss any five (5) methods of locating features in maps. (10 marks)
4. Discuss the factors which influence the site, distribution and pattern of settlement. (20 marks)
5. Define the term GIS and discuss its relevance in mapwork.. (20 marks)
6. Write explanatory notes on the following:
 - a. Sketch Map (3 marks)
 - b. Large Scale (5 marks)
 - c. Short Path Matrix (6 marks)
 - d. Stream Order (6 marks)



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FIRST YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF EDUCATION WITH INFORMATION TECHNOLOGY (CITY CAMPUS - SCHOOL BASED)

NGE 105: FUNDAMENTALS OF PHYSICAL GEOGRAPHY

Date: 29th April, 2013

Time: 2.00 – 4.00pm

INSTRUCTIONS:

- ♦ Answer QUESTION ONE and any other TWO questions.
- ♦ Sketch maps and diagrams should be used whenever appropriate.

NGE 105 FUNDAMENTALS OF PHYSICAL GEOGRAPHY

1. a) Describe the Earth's crust and the asthenosphere. (6 marks)
b) Explain the significance of erosion in shaping the Earth's landscape. (8 marks)
c) Describe the structure of the atmosphere. (8 marks)
d) Explain the concept of Ecosystem. (8 marks)
2. Discuss the view that the Earth's thermal history is constant. (20 marks)
3. Define the plate boundaries and describe their associated features. (20 marks)
4. Describe the weathering process and its associated products. (20 marks)
5. Citing examples in Kenya, explain the economic significance of sedimentary rocks. (20 marks)
6. Discuss the world's major climates according to Koppen's Classification Scheme. (20 marks)



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NGE 106: SPATIAL ORGANIZATION

Date: 2nd May, 2013

Time: 2.00 – 4.00pm

INSTRUCTIONS:

- ◆ Answer QUESTION ONE and any other TWO questions.
- ◆ Sketch maps and diagrams should be used whenever appropriate.



NGE 106 SPATIAL ORGANIZATION

1. a) Explain the four (4) phases in the development of Geography as a science. (8 marks)
 b) Examine the major sub-divisions of human Geography. (8 marks)
 c) Discuss the extent to which the gravity model is applicable in explaining spatial flow patterns in Kenya. (14 marks)
2. “Although the shortest distance between an origin and a destination is the straight line between them, in practice, people rarely follow such path”. Discuss. (20 marks)
3. “Profit maximization remains the most useful simplifying assumption for the spatial analysis of farming patterns”. Discuss. (20 marks)
4. Discuss the strenghts and weaknesses of of Alfred Weber’s theory of industrial location. (20 marks)
5. Discuss the Ullman Harris Multiple Nuclei model and show its relevance to Kisumu City. (20 marks)
6. Citing examples in East Africa, examine the the factors that influence the growth of urban settlements. (20 marks)



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NGE 201: GEOGRAPHIC THOUGHT

Date: 21st July, 2013

Time: 11.00 a.m. – 1.00 p.m.

INSTRUCTIONS:

- ♦ Answer Question ONE and any other TWO (2) questions.
- ♦ Sketch maps and diagrams should be used whenever appropriate.



NGE 201 GEOGRAPHIC THOUGHT

1. a) Discuss the view that geography is not an island discipline. (10 marks)
b) Explain the pros and cons of the geographical paradigm of environmental determinism. (10 marks)
c) Explain the effects of the quantitative revolution. (10 marks)
2. Justify the existence of the discipline geography between 250 BC and 950 BC. (20 marks)
3. Outline the contribution of the following scholars to the advancement of geographic knowledge.
 - a. Friedrich Ratzel
 - b. Erastosthenes
 - c. William Morris Davis
 - d. Claudius Ptolemy(20 marks)
4. Discuss the contributions of Alexander Von Humboldt and Karl Ritter to the discipline of geography. (20 marks)
5. Discuss the concept of “regional geography” as postulated by the French School of thought. (20 marks)
6. Citing specific examples, examine the role of geography in environmental management. (20 marks)



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NGE 201: GEOGRAPHIC THOUGHT

Date: 29th August, 2013

Time: 2.00 – 4.00 p.m.

INSTRUCTIONS:

- ◆ Answer Question ONE and any other TWO (2) questions.
- ◆ Sketch maps and diagrams should be used whenever appropriate.



NGE 201 GEOGRAPHIC THOUGHT

1. a) Discuss the Relationship between geography and other human disciplines. (10 marks)
b) Explain Charles Darwin's contribution to geography. (10 marks)
c) Explain the effects of the quantitative revolution on geography. (14 marks)
2. Examine the evolutionary phases in the discipline of geography. (20 marks)
3. Discuss the contributions of the Germans to the discipline of geography. (20 marks)
4. Write explanatory notes on any four (4) of the following:
a) Paul Vidal (5 marks)
b) William Morris Davis (5 marks)
c) Halford Mackinder (5 marks)
d) Ellen Churchill Semple (5 marks)
e) Al Idrisi (5 marks)
5. Assess the relevance of the concept of Environmental Determinism to socio-economic development in Africa. (20 marks)
6. Citing examples, examine the importance of geography in the promotion of environmental awareness. (20 marks)



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NGE 202: CLIMATOLOGY

Date: 26th August, 2013

Time: 2.00 – 4.00 p.m.

INSTRUCTIONS:

- ♦ Answer Question ONE and any other TWO (2) questions.
- ♦ Sketch maps and diagrams should be used whenever appropriate.



NGE 202 CLIMATOLOGY

1. a) Describe the associated weather conditions of the five (5) major air masses. (10 marks)
 b) Explain the concept of “Green House” effect. (10 marks)
 c) Explain any five (5) causes of climate change. (10 marks)
2. Discuss the processes that determine the amount of radiant energy received on the earth surface. (20 marks)
3. Explain the rain drop formation theories. (20 marks)
4. Explain the hydrological cycle and its effect on climate. (20 marks)
5. Explain the factors that influence the formation of tropical cyclones. (20 marks)
6. “Desertification is a natural phenomenon”. Discuss. (20 marks)



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NGE 202: CLIMATOLOGY

Date: 27th July, 2013

Time: 8.00 – 10.00 a.m.

INSTRUCTIONS:

- ♦ Answer Question ONE and any other TWO (2) questions.
- ♦ Sketch maps and diagrams should be used whenever appropriate.



NGE 202 CLIMATOLOGY

- a) Describe the vertical structure of the atmosphere. (10 marks)
 - b) Describe the five (5) major air masses. (10 marks)
 - c) Describe Koppen's major climate classification scheme. (10 marks)
2. Discuss the effect of green house gases in global warming. (20 marks)
3. "The atmosphere is an energy filter in two directions". Discuss. (20 marks)
4. Discuss the view that the hydrological cycle is the nature's chief water distillation plant. (20 marks)
5. Discuss any three (3) forces which govern horizontal wind movement. (20 marks)
6. "Anthropogenic factors are the major cause of climate change". Discuss. (20 marks)



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NGE 206: EAST AFRICAN ENVIRONMENT

Date: 19th July, 2013

Time: 11.00 a.m.- 1.00 p.m.

INSTRUCTIONS:

- ♦ Answer Question ONE and any other TWO (2) questions.
- ♦ Sketch maps and diagrams should be used whenever appropriate.



NGE 206 EAST AFRICAN ENVIRONMENT

1. a) Explain the nature of the subject matter of geography. (10 marks)

 b) Explain the special contribution expected from the Geographer in East Africa. (20 marks)
2. Explain the importance of the following with regard to the East African environment.
 - a. Geographical location (8 marks)
 - b. Regionalization (4 marks)
 - c. Environments (4 marks)
 - d. Ecosystems (4 marks)
3. Explain the key concepts of geography and show their relevance to economic development of East Africa. (20 marks)
4. Explain the major physical environmental constraints on development of East Africa. (20 marks)
5. Evaluate the nature, patterns and problems of urbanization in East Africa. (20 marks)
6. Explain the usefulness of physiographic approach to land use planning in Kenya. (20 marks)



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NGE 207: ENVIRONMENT AND DEVELOPMENT IN AFRICA

Date: 18th July, 2013

Time: 11.00 a.m. – 1.00 p.m.

INSTRUCTIONS:

- ◆ Answer Question ONE and any other TWO (2) questions.
- ◆ Sketch maps and diagrams should be used whenever appropriate.



NGE 207 ENVIRONMENT AND DEVELOPMENT IN AFRICA

1. a) Briefly describe the physiography of Africa. (10 marks)
b) Examine the draw backs in Africa's development. (10 marks)
c) Explain the variations in climatic conditions in Africa. (10 marks)
2. Discuss the relevance of the concept of Sustainable Development in Africa. (20 marks)
3. Define the term Environmental Change and discuss its impact in Africa. (20 marks)
4. Evaluate the impact of the rapid population growth in Africa on the environment and development. (20 marks)
5. "Climate and soils of Africa are to blame for the pervasive food insecurity in the continent". Discuss. (20 marks)
6. Discuss the impact of energy production on the environment and development in Africa. (20 marks)



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NGE 208: GEOMORPHOLOGY

Date: 23rd July, 2013

Time: 2.30 – 4.30 p.m.

INSTRUCTIONS:

- ♦ Answer Question ONE and any other TWO (2) questions.
- ♦ Sketch maps and diagrams should be used whenever appropriate.



NGE 208 GEOMORPHOLOGY

1. a) Critically examine the meanings of Lyell's uniformity of state and of rate. (6 marks)
 b) Describe process - response systems in geomorphology. (6 marks)
 c) Describe any two (2) processes of chemical weathering and their resultant features. (8 marks)
 d) Discuss the simulation approach in studying the nature of landscape change through time. (10 marks)
2. Describe equilibrium and evolution of landforms. (20 marks)
3. Describe the types of rapid flowage mass movement and give their diagnostic features. (20 marks)
4. Discuss the relationship between geomorphology and other disciplines. (20 marks)
5. Describe depositional fluvial landforms and give examples in Kenya. (20 marks)
6. Describe constructional coastal features and explain their significance to tourism in Kenya. (20 marks)



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NGE 209: AGROFORESTRY SYSTEMS

Date: 24th July, 2013

Time: 8.30 – 10.30 a.m.

INSTRUCTIONS:

- ♦ Answer Question ONE and any other TWO (2) questions.
- ♦ Sketch maps and diagrams should be used whenever appropriate.



NGE 209 AGROFORESTRY SYSTEMS

1. a) Trace the history of agroforestry. (8 marks)

b) Describe four (4) bases on which agroforestry systems are categorized. (10 marks)

c) "The disparity between farmers' perception and the technical experts' preconceptions of agroforestry has been a set back to the global development of agroforestry". Discuss. (12 marks)
2. a) Differentiate between the three (3) major tropical agroforestry regions. (10 marks)

b) Assess the relevance of the Taungya System to agroforestry in Kenya. (10 marks)
3. Discuss agroecological zonation and socio-economic agroforestry systems' classification criteria. (20 marks)
4. Examine Shifting Cultivation and Alley Cropping agroforestry subsystems in the developing countries. (20 marks)
5. a) Compare and contrast agroforestry in the tropical and temperate zones. (8 marks)

b) Discuss Intercropping under hardwood species and Silvopastoral agroforestry systems in the developed countries. (12 marks)
6. Discuss three (3) socio-cultural factors that influence the adoption of agroforestry practices. (20 marks)



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MANAGEMENT AND BACHELOR OF EDUCATION
WITH INFORMATION TECHNOLOGY
(HOMA BAY CAMPUS - SCHOOL-BASED)**

NGE 306: URBAN GEOGRAPHY

Date: 27th July, 2013

Time: 8.00 – 10.00 a.m.

INSTRUCTIONS

- ♦ Answer Question ONE (COMPULSORY) and any other TWO questions.
- ♦ Sketch maps and diagrams should be used whenever appropriate.



NGE 306 URBAN GEOGRAPHY

1.
 - a. Differentiate between urban system and urban as a system. (6 marks)
 - b. Giving examples, describe the secondary and tertiary sectors of urban areas. (6 marks)
 - c. Explain the main objectives of urban zoning and the factors to be considered in developing an urban zoning plan. (12 marks)
 - d. Explain any three (3) factors that affect urban development. (6 marks)
2. a) Explain the five (5) approaches that have historically characterized urban geography since the 1950s to date. (10 marks)
b) Describe the distinct periods of evolution of modern urban centres. (10 marks)
3. Discuss the categories of urban problems that affect many cities of the developing countries. (20 marks)
4. Discuss the factors that have hampered the process of zoning for urban areas in Kenya. (20 marks)
5. a) Examine the characteristics of an ideal community inhabiting a modern city. (10 marks)
b) Describe the generational models used for urban resettlement process. (10 marks)
6. Using illustrations, examine the Concentric Zone and the Hoyt models of describing internal structures of cities. (20 marks)



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NGE 317: RESEARCH METHODS

Date: 15th July, 2013

Time: 2.00 – 4.00 p.m.

INSTRUCTIONS:

- ♦ Answer Question ONE and any other TWO (2) questions.
- ♦ Sketch maps and diagrams should be used whenever appropriate.



NGE 317 RESEARCH METHODS

1.
 - a. Explain the importance of research in geography. (10 marks)
 - b. Write explanatory notes on the following:
 - i. Tools of Research (10 marks)
 - ii. Evaluative Research (5 marks)
 - iii. Primary Sources of Data (5 marks)
2. “The research process is cyclical”. Discuss. (20 marks)
3.
 - a. Examine the significance of sampling in geographic studies. (10 marks)
 - b. Evaluate the role of Literature Review in geographic research. (10 marks)
4. Describe the guidelines that should be followed when designing an effective questionnaire. (20 marks)
5. Discuss the characteristics of qualitative research. (20 marks)
6. Distinguish between descriptive and inferential statistics. (20 marks)



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NGE 318: BIOGEOGRAPHY

Date: 15th July, 2013

Time: 9.00 – 11.00 a.m.

INSTRUCTIONS:

- ♦ Answer Question ONE and any other TWO (2) questions.
- ♦ Sketch maps and diagrams should be used whenever appropriate.



NGE 318 BIOGEOGRAPHY

1. a) Explain the concept of Island Biogeography. (10 marks)
b) Examine the significance of studying biogeography. (10 marks)
c) Explain any five (5) forms of human environment interaction. (10 marks)
2. Examine the structure and function of an ecosystem. (20 marks)
3. Discuss the causes of biodiversity degradation and loss and suggest possible conservation measures. (20 marks)
4. Examine the factors affecting ecosystems. (20 marks)
5. Discuss the causes of the distribution of life forms in the ecosphere. (20 marks)
6. a) Explain the types of climax communities. (10 marks)
b) Discuss any five (5) adaptive strategies of biota in savanna ecosystems. (10 marks)